

Undergraduate educational activity

Designing with GaN – miniaturization and heating of high frequency converter

Purpose: development of educational demo board with the possibility of observing by students the heating of the system elements with changes in parameters such as frequency, load, voltage and gain.

Assumptions:

- topology: synchronous DC-DC buck converter
- top-cooling GaN discrete transistors e.g. EPC2367
- control with the possibility of fixing the switching frequency (e.g. 500 kHz, 1MHz, 1.5MHz, 2MHz) and gain: 40/12V, 48/5V and 48/1V.
- heatsink and controlled fan (e.g. 3 speeds: 0, 1, 2)
- temperature sensors under transistors (e.g. TMP118)
- control system based on a microcontroller (e.g. STM)
- measurements on the board: input and output voltage
- displaying the temperature values of the elements and system parameters

Electrical assumptions:

- U_{in} : 48 V
- U_{out} : adjustable
- L : optimal for $P_{out}=100\text{ W}$, $f=500\text{kHz}$, 48/12V
- f : adjustable
- Transistors: GaN EPC module EPC2367
- Temperature sensor
- Controller and measurement management, parameter setting and display of results – microcontroller

Stages:

1. Component selection – GaN module, passives LC, controller, sensors, data display.
2. Altium Designer schematic.
3. Altium Designer PCB design of buck converter power converter.
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5. PCB production.
6. Verification and assembling.
7. Laboratory tests.